KAI-PO WEI

Ann Arbor, MI | kaipowei@umich.edu | github.com/kaipowei | linkedin.com/in/kaipowei-umich

EDUCATION

University of Michigan — **Ann Arbor, MI**
Master of Science in Robotics — *May 2028*
Coursework: Mathematics for Robotics, Robotic Systems Laboratory

National Taipei University of Technology — **Taipei, Taiwan**
Bachelor of Science in Vehicle Engineering — *Jan 2026*
GPA: 3.61/4.0
Coursework: Vehicle Dynamics, Automatic Control, Statics, Mechanics of Materials

TECHNICAL SKILLS

Programming: Python, C++, CUDA, Git

Robotics / Autonomy: ROS2 (rclpy, rclcpp), Motion Planning (Hybrid-A*, Pure Pursuit, Stanley, MPC), State Estimation (EKF, ICP), Gazebo, Rviz, PCL

ML / Computer Vision: PyTorch, OpenCV, MONAI, Ultralytics YOLO, MATLAB, Simulink

Vehicle Dynamics & CAE: CarSim, BikeSim, Tire Modeling, SolidWorks, Altair HyperWorks

PROJECT EXPERIENCE

Friction-Aware Motion Planning & Perception Stack for Autonomous Vehicles — **Taipei, Taiwan**
Autonomous System Researcher, [Planner Repo], [Perception Repo] — *Jun 2026 – Aug 2026*

- Engineered a custom C++17/CUDA voxel-grid downsampler and RANSAC/Euclidean-clustering pipeline for real-time LiDAR perception.
- Benchmarked a 4.5x GPU speedup at 1M points and eliminated 5 of 7 false-positive detections with zero loss of true positives.
- Modeled a dynamic bicycle vehicle with a friction-circle-saturated tire model and designed an EKF estimating road friction and sideslip live from noisy IMU data, tracking a sudden dry-to-ice transition within 0.015 of the true value in 0.02s.
- Designed a friction-aware Hybrid-A* planner and nonlinear MPC/Pure-Pursuit controller bounded by the tire friction circle, validated via Pytest; raised controller success from 22/36 to 36/36 conditions across a 72-run sweep, cutting worst-case tracking error from 20.4m to 0.34m.
- Fused a LiDAR/IMU EKF for localization (17% error reduction) and a live road-friction/sideslip EKF (converges within 0.015μ in 0.02s) into a closed-loop ROS2 pipeline, reaching a 6m goal autonomously in 13.5s within a Gazebo simulation.

AI CUP 2025: Cardiac CT Segmentation & Detection — **Taipei, Taiwan**
Perception Researcher, Advisor: Prof. Tung-Ju Hsieh — *Sep 2025 – Jan 2026*

- Developed a MONAI UNet3D volumetric segmentation pipeline with automated intensity clipping and spatial resampling to 128x128x128 ROIs. Optimized using AdamW and Dice Cross-Entropy loss to achieve a 0.7439 leaderboard score.
- Trained a YOLOv12 object detection model on a 2,700+ frame dataset. Systematically tuned inference hyperparameters including NMS IoU (0.65) and confidence thresholds, reaching a 0.9396 evaluation score.

Scooter Frame Fatigue Analysis & Optimization — **Taipei, Taiwan**
Team Leader & Primary CAE Engineer, Structural & Optimization Lab, NTUT — *Sep 2023 – Jul 2024*

- Led a 4-person team to design a multi-accelerometer test rig and derived a geometric sensor-fusion algorithm converting raw accelerometer signals into center-of-gravity motion, validated per CNS 15819-5 durability regulations.
- Built a SOLIDWORKS/HyperMesh FEA model and validated it against real road-test data via a bikeSIM multibody-dynamics simulation across 4 road conditions; used PSD fatigue analysis to optimize frame thickness, driving a 339% improvement in fatigue life (3,093 → 13,582 cycles) to exceed a 100,000km durability target.

WORK EXPERIENCE

Foxconn, Center of Commercial Electric Vehicles — **New Taipei City, Taiwan**
Vehicle Integration Intern — *Jul 2023 – Aug 2023*

- Contributed cockpit-ergonomics proposals for the full-electric commercial van ‘Model N’ (2 adopted).
- Collaborated cross-functionally with ADAS, Chassis, and Human-Factors teams.

Taipei Tech Racing (Formula Student) — **Taipei, Taiwan**
Chassis Team Member — *Oct 2022 – Jul 2023*

- Designed drivetrain CAD and suspension geometry for the TTR7 race car (2nd place, Formula Student Taiwan 2023); achieved a 31% cost reduction and 16% weight savings.